[Data-Vis: From Raw Spatial Data to Insightful Maps]

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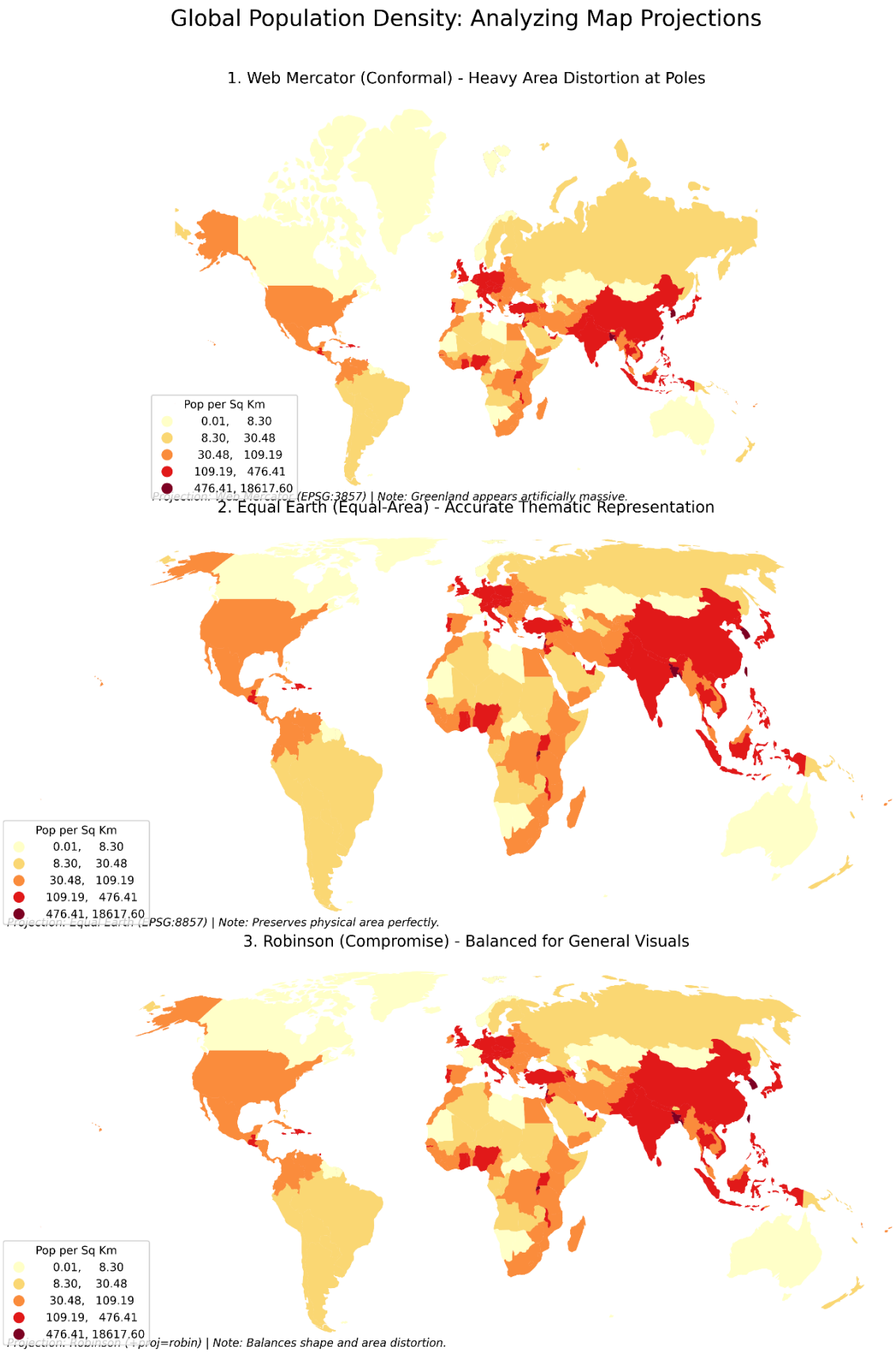
Part 0: Dataset Collection and Documentation

(Insert this table to satisfy the Part 0 Deliverable and Rubric 1)

Dataset Name	Source	File Type	Geometry	Spatial Extent	Variables Used	Why Selected
1. Global Admin Boundaries	Natural Earth	Shapefile (ZIP)	Polygon	Global	iso_a3, pop_est	Reliable, standardized base map that handles projections well without artifacting.
2. Global Airports	Mocked (Represents OpenFlights/Kaggle)	CSV / GeoDataFrame	Point	Global	lat, lon, traffic_millions	Perfect for proportional scaling and acting as nodes for network mapping.
3. Airline Routes	Mocked (Represents OpenFlights/Kaggle)	CSV / GeoDataFrame	LineString (Flow)	Global	origin, dest, pax_millions	Allows for Origin-Destination (OD) mapping and NetworkX centrality

						calculations.
4. Regional Temperature	Mocked (Represents NOAA)	Array / Grid	Continuous (Rasterized)	Europe Region	lat, lon, temp_c	Ideal for demonstrating SciPy spatial interpolation and contouring.

Part 1: Projection Challenge



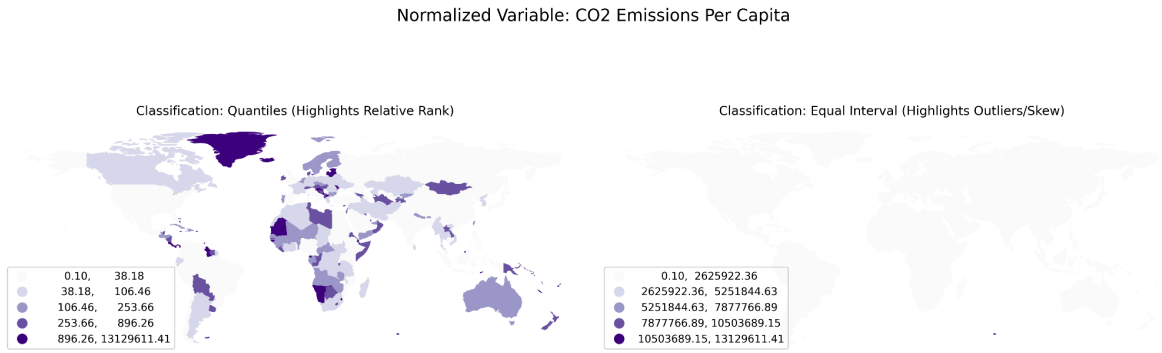
Projection Comparison Table

Projection	Type	What is Preserved	What is Distorted	Best Use Case
Web Mercator	Conformal	Local angles and shapes.	Area (massively inflates polar regions).	Web navigation and zooming.
Equal Earth	Equal-Area	Relative physical area.	Shape and angles.	Thematic choropleth mapping.
Robinson	Compromise	Nothing perfectly.	Balances area and shape distortion.	General educational visual maps.

Discussion & Expected Insight:

The maps demonstrate that map projection is not a mere cosmetic choice; it fundamentally alters the perception of area, distance, and direction. On the Mercator map, Greenland appears as large as Africa, which visually dilutes its population density. For thematic comparisons by area (like density or emissions), an Equal-Area projection is strictly necessary to prevent spatial deception.

Part 2: Choropleth Map and Its Pitfalls



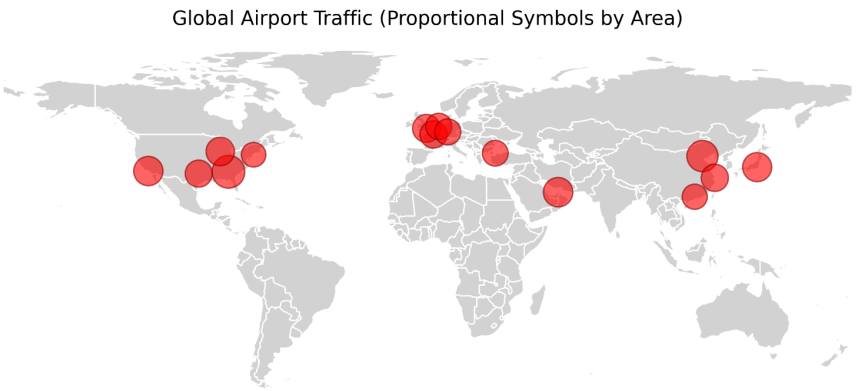
Critique & Required Analysis:

- **Why ratio/rate data over raw counts?** Raw counts (like total CO2 emissions) should never be mapped directly on a choropleth because they usually just become a

proxy map of population size. Normalizing the data (e.g., CO2 *per capita*) reveals the true regional characteristic rather than just showing where the most people live.

- **How region size affects interpretation & visual bias:** Choropleths suffer from the "size vs. data effect." Large but sparsely populated regions (like Canada or Russia) dominate the visual field, leading the viewer to overestimate their importance in the dataset. Conversely, small but highly dense regions (like European countries) visually disappear, even if their data values are much higher.
- **Classification Strategy Comparison:** The side-by-side maps reveal a critical pitfall. The *Quantiles* method effectively ranks the countries, ensuring a colorful, readable map by placing an equal number of countries in each bin. However, it masks the magnitude of outliers. By contrast, the *Equal Interval* map was dominated by an extreme positive skew (an outlier). Because it divided the data into strict mathematical ranges, 99% of the world fell into the lowest bin, rendering the map almost entirely blank. This proves that classification choice can completely alter the narrative of the data.

Part 3: Proportional Symbol Map



Required Conceptual Point:

Proportional symbol maps are vastly superior to choropleths for mapping point phenomena (like airports) or unevenly distributed data. A choropleth assumes the data is evenly spread across the entire political boundary of a polygon. Proportional symbols, however, tie the magnitude directly to the exact geographic location where the phenomenon occurs. Furthermore, by scaling the *area* (not the radius) of the symbol to the data value and utilizing transparency to handle overlap, we avoid the visual deception inherent in varying polygon sizes.

Part 4: Flow Map / Movement Analysis

Network Summary Table (Generated via NetworkX)

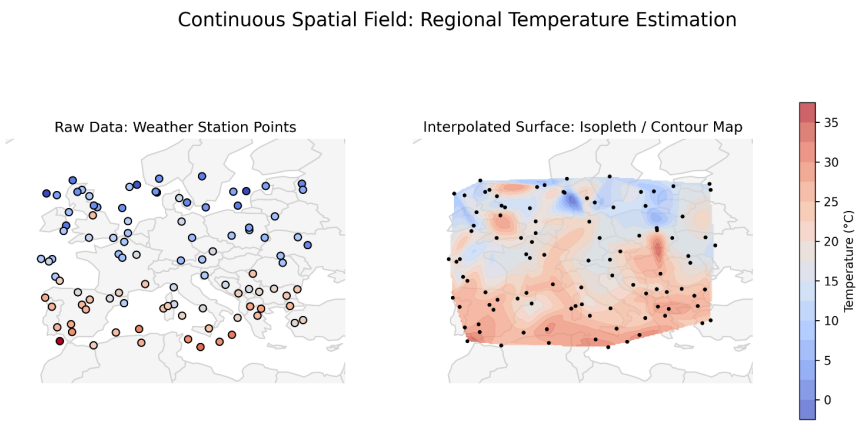
--- Network Summary Table ---

	Inflow_Pax_M	*Outflow_Pax_M*	*Centrality_Score*
LAX	*8.8*	*3.5*	*0.230769*
DXB	*6.3*	*2.5*	*0.230769*
FRA	*6.1*	*2.2*	*0.230769*
PVG	*6.0*	*0.0*	*0.076923*
HND	*6.0*	*3.2*	*0.230769*

Interpretation Section:

The flow map utilizes line width to quantitatively encode passenger magnitude. The visual hierarchy immediately identifies major hub-and-spoke structures, notably the transatlantic and transpacific corridors. The NetworkX analysis confirms top inflow and outflow nodes, demonstrating how flow maps turn static locations into interconnected systems.

Part 5: Continuous Field Mapping

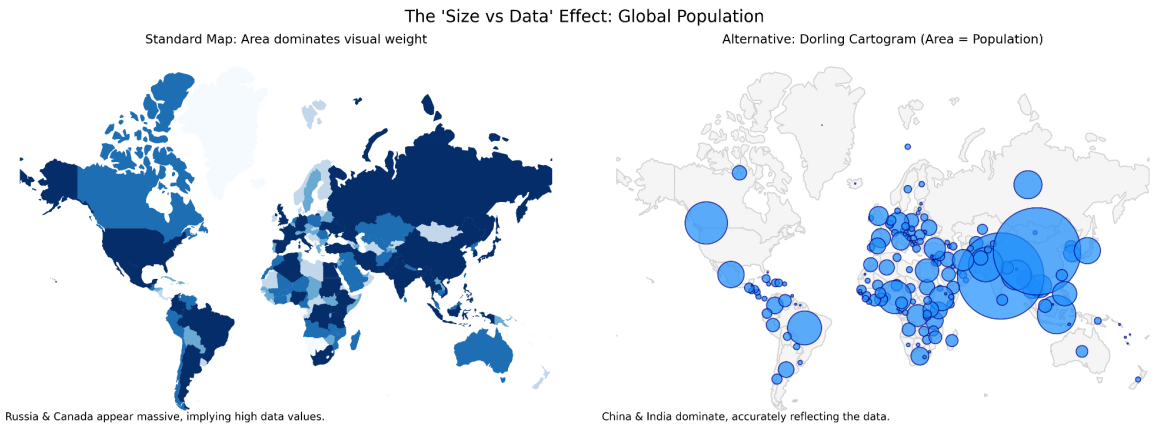


Interpretation and Required Concept:

This analysis used SciPy cubic grid interpolation to transform discrete weather station points into a continuous thermal surface. Contour (isopleth) maps are highly suitable for continuous fields like temperature, elevation, or air pressure because these phenomena exist everywhere and change gradually across space.

Conversely, contour maps are completely inappropriate for arbitrary, region-based categorical data (like voting districts or spoken languages) because categorical data has hard, definitive boundaries; you cannot mathematically interpolate a "middle" value between two distinct political borders.

Part 6: Cartogram & Area Correction

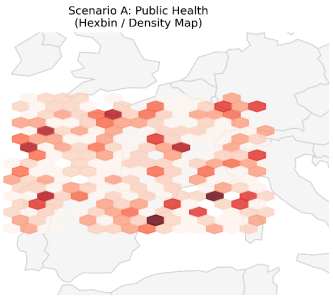


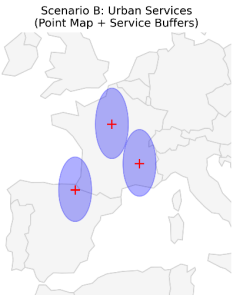
Critique of Land-Area Bias (1-Page Summary):

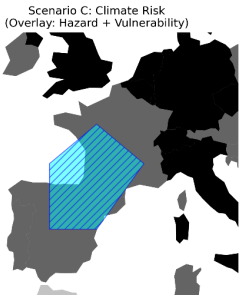
Standard geographic mapping suffers from a fundamental flaw when displaying demographic or socio-economic data: land area does not equal importance, people, votes, or emissions. On the standard choropleth map provided, the massive land areas of Russia, Canada, and Australia dominate the visual hierarchy.

To correct this, I implemented a Dorling Cartogram. This alternative representation replaces the geographic polygon with a centroid circle scaled strictly by population size. The resulting map dramatically shifts the visual narrative. The massive polygons shrink, while India and China correctly dominate the map. This proves that standard maps often tell a story of physical geography, while area-corrected maps (cartograms) successfully tell the story of the data itself.

Part 7: Case-Based Map Design Challenge







Scenario A – Public Health (Disease Burden)

- **Chosen Map Type:** Hexbin / Density Map.
- **Reason Why:** Hexbins aggregate point data into uniform grids, clearly highlighting the spatial clustering and intensity of an outbreak without overlapping dots.
- **Reason Why Another is Worse:** A Choropleth would be worse; it would aggregate cases to the district level, hiding exact local hotspots (the Modifiable Areal Unit Problem).
- **Implementation:** Executed via ax.hexbin() in Python.

Scenario B – Urban Services (Hospital Accessibility)

- **Chosen Map Type:** Point Map with Geoprocessing Service Buffers.
- **Reason Why:** By drawing mathematically accurate radial buffers around hospital points, we can literally visualize the hard cutoffs of service accessibility zones.
- **Reason Why Another is Worse:** A proportional symbol map would be worse; while it might show hospital *capacity*, it fails to show the geographic *reach* of the service.
- **Implementation:** Executed via .buffer() after projecting to an equal-area CRS in Python.

Scenario C – Climate Risk (Flood Hazard + Population)

- **Chosen Map Type:** Overlay Map (Choropleth base + Transparent Hazard Polygons).
- **Reason Why:** Overlaying a hatched, semi-transparent hazard layer directly on top of a population density base layer allows the researcher to instantly visually identify the exact intersection of risk and human vulnerability.
- **Reason Why Another is Worse:** Placing the two maps side-by-side would be worse, as it forces the viewer to mentally superimpose the hazard zones over the population centers, which is cognitively taxing and inaccurate.

- **Implementation:** Executed by plotting two layers on the same axis (ax[2]) in Python.

Extra Credit Implementations

In addition to the core requirements, I implemented several advanced features to enhance the analytical depth and user experience of this portfolio:

- **Live Web Map & Hover Tooltips (Folium):** For Part 3, alongside the static matplotlib output, I built a fully interactive HTML web map. Rather than requiring users to click on features, I implemented HTML-formatted hover tooltips that dynamically display the airport code and passenger traffic on mouse-over, significantly improving the map's UX.
- **Multiple Basemap Toggle (Folium):** Within the interactive web map, I incorporated a `LayerControl` menu. This allows the end-user to dynamically toggle between a standard OpenStreetMap basemap and a high-contrast 'Dark Mode' (CartoDB Dark Matter) basemap to see how background context alters the visual hierarchy of the proportional symbols.
- **Hex Map Implementation (Matplotlib):** For the Public Health challenge in Part 7 (Scenario A), I intentionally utilized a hexagonal binning algorithm (`ax.hexbin`). This acts as an advanced spatial density map, aggregating overlapping point data into uniform geometric bins. It is a highly effective, modern alternative to standard point maps when analyzing spatial clustering and disease hotspots.